Research with AI Agents

What Works, What It Costs, and What an Institute Needs

Lukas Theissinger

October 30, 2026

Outline

OpenAI and Agent Usage

Owning Intelligence and Infrastructure at Lamarr

Staying Up to Date with AI

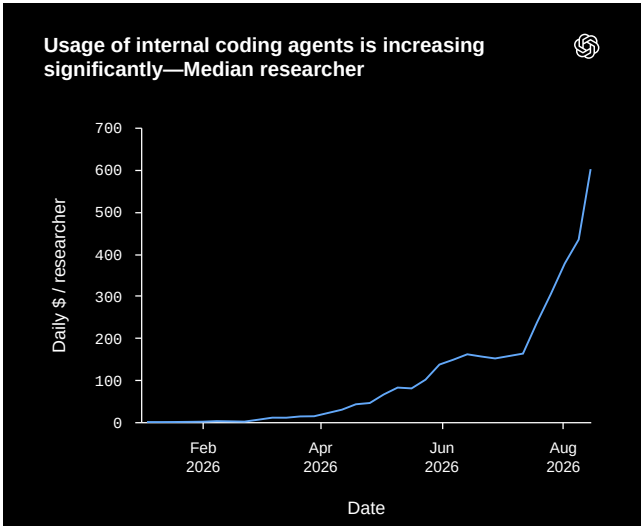
Outline

OpenAI and Agent Usage

Owning Intelligence and Infrastructure at Lamarr

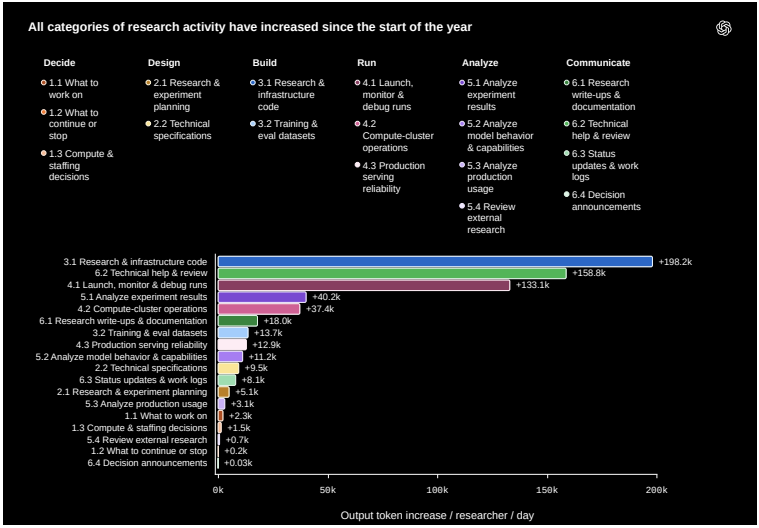
Staying Up to Date with AI

FOMO Headline: OpenAI Agent Usage



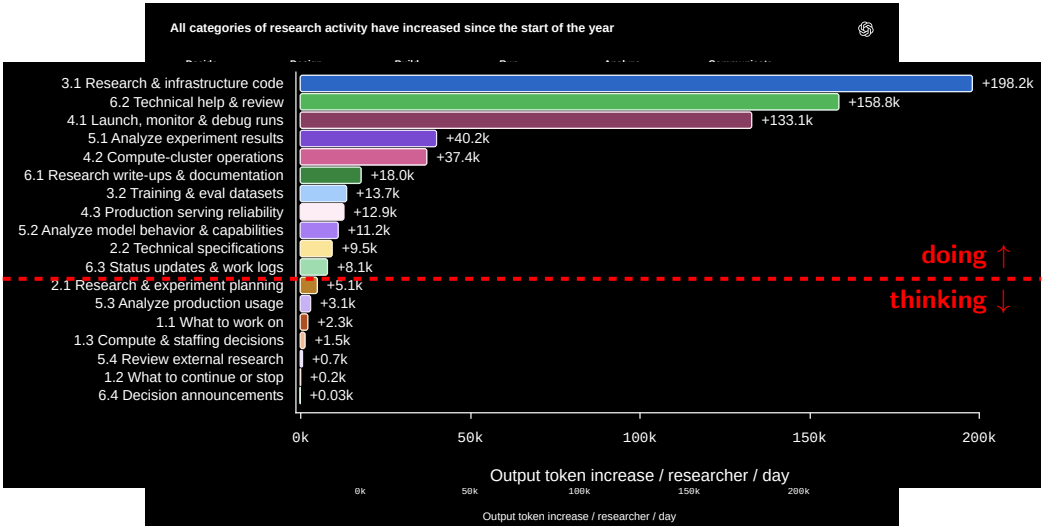
Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported).

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, "Research acceleration" (self-reported).

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.

Agents took over the *doing*: code, runs, technical help.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.

Agents took over the *doing*: code, runs, technical help.

Frequent, repetitive work grew most; rare decisions grew least.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.

Agents took over the *doing*: code, runs, technical help.

Frequent, repetitive work grew most; rare decisions grew least.

Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests	Dettmers says
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.	Use agents or be left behind.
Agents took over the *doing*: code, runs, technical help.	
Frequent, repetitive work grew most; rare decisions grew least.	
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests	**Dettmers says**
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.	Use agents or be left behind.
Agents took over the *doing*: code, runs, technical help.	"More than 90% of code and text should be written by agents."
Frequent, repetitive work grew most; rare decisions grew least.	
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests	Dettmers says
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.	Use agents or be left behind.
Agents took over the *doing*: code, runs, technical help.	"More than 90% of code and text should be written by agents."
Frequent, repetitive work grew most; rare decisions grew least.	Automate only if how often $\times$ time saved $>$ time to build.
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests	**Dettmers says**
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.	Use agents or be left behind.
Agents took over the *doing*: code, runs, technical help.	"More than 90% of code and text should be written by agents."
Frequent, repetitive work grew most; rare decisions grew least.	Automate only if how often $\times$ time saved $>$ time to build.
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	Agent use is a *skill*: not everything can or should be automated (degree of automation).

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every week:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every week**

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every week:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every week**

2. **Automate as much as possible.**
   a) Task solving: an agent solves every task, with code and plots
   b) Slide creation: an agent turns the solutions into the class slides

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every week:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every week**

2. **Automate as much as possible.**
   a) Task solving: an agent solves every task, with code and plots
   b) Slide creation: an agent turns the solutions into the class slides

3. **Left to do:** understand the solutions, make small updates to the slides.
   - class stays 2 h, preparation drops from 6 h to 2 h: **2 h + 2 h = 4 h every week**

Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every week:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every week**

2. **Automate as much as possible.**
   a) Task solving: an agent solves every task, with code and plots
   b) Slide creation: an agent turns the solutions into the class slides

3. **Left to do:** understand the solutions, make small updates to the slides.
   - class stays 2 h, preparation drops from 6 h to 2 h: **2 h + 2 h = 4 h every week**

4. **Try it yourself:** `github.com/Newida/AutomatingExercises`

```
$ auto_exercises sheet.pdf
```

Outline

OpenAI and Agent Usage

Owning Intelligence and Infrastructure at Lamarr

Staying Up to Date with AI

Small Labs Can Compete

What It Costs

What an Institute Needs

Takeaways

Outline

OpenAI and Agent Usage

Owning Intelligence and Infrastructure at Lamarr

Staying Up to Date with AI

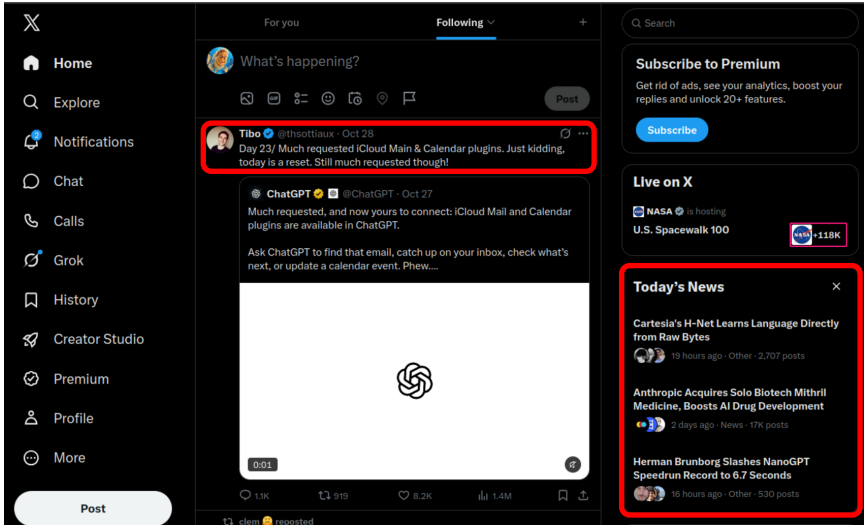
How to Stay Up to Date

1. **Make an X (Twitter) account.**
   Many people from the frontier labs post there first, often daily.

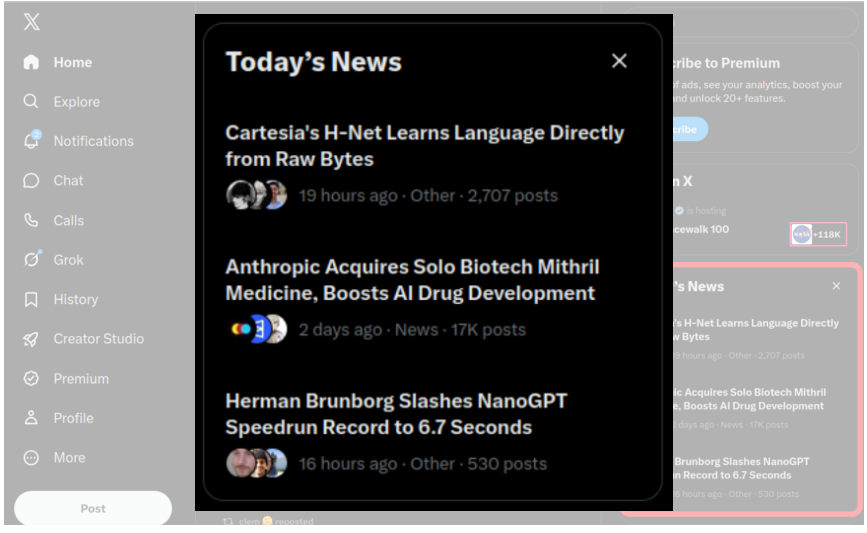
2. **Read "Today's News" on the home page.**
   It sums up the biggest updates of the day.

3. **Follow the right people.**
   Don't be fooled by clickbait.

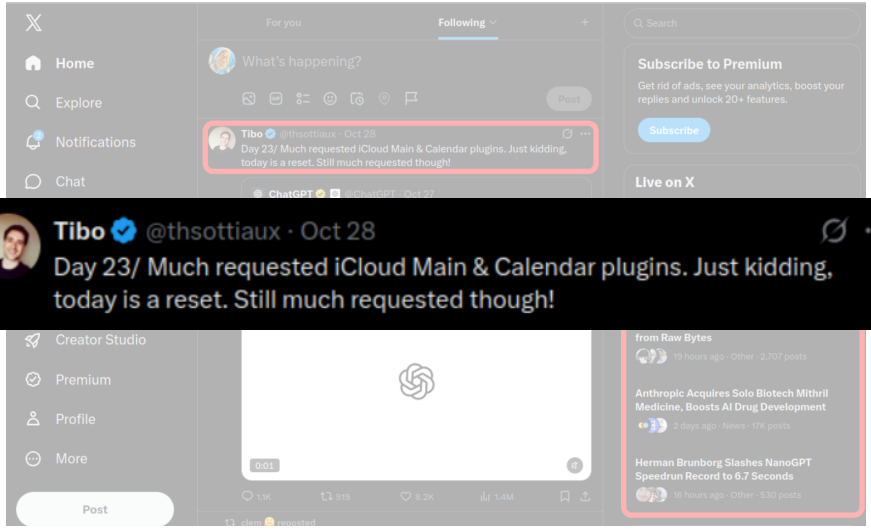
Read Today's News



Read Today's News



Read Today's News



Tibo @thsottiaux · Oct 28

Day 23/ Much requested iCloud Main & Calendar plugins. Just kidding, today is a reset. Still much requested though!

Who to Follow and Why

Who	Handle	Why
Andrej Karpathy	@karpathy	Clear explanations of LLMs and how to work with them
Thibault Sottiaux (tibo)	@thsottiaux	Leads OpenAI Codex; news on Codex releases and usage limits
Boris Cherny	@bcherny	Created Claude Code; how Anthropic uses its own agents
Tim Dettmers	@Tim_Dettmers	Agents in academic research; open models on small budgets
Victor Taelin	@VictorTaelin	Tests frontier models on hard coding problems
Felix Rieseberg	@felixrieseberg	Builds Claude's desktop apps at Anthropic; early product news

A few of the 251 accounts I follow, all of this kind.

A Recent Headline You Might Have Missed



A Recent Headline You Might Have Missed

